NCAA Basketball Predictor

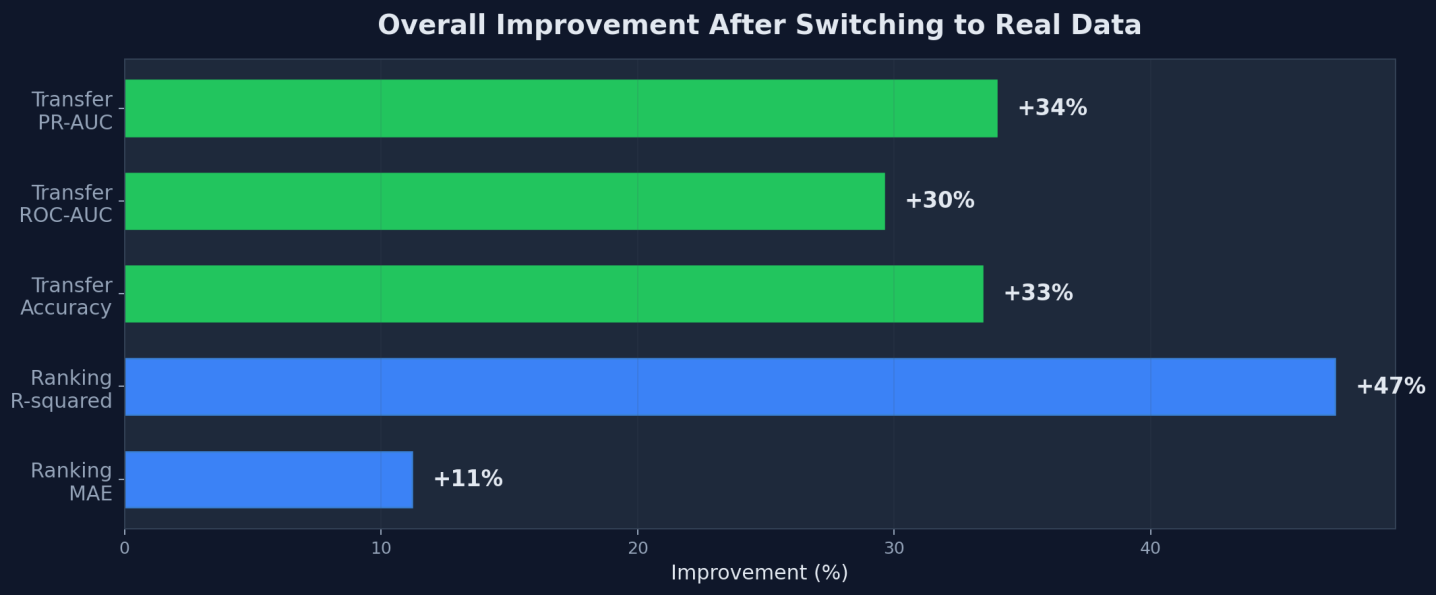
Model Performance Report | Real Data vs Synthetic Baseline

Executive Summary

After replacing synthetic training data with real BartTorvik game data, both models showed dramatic accuracy improvements:

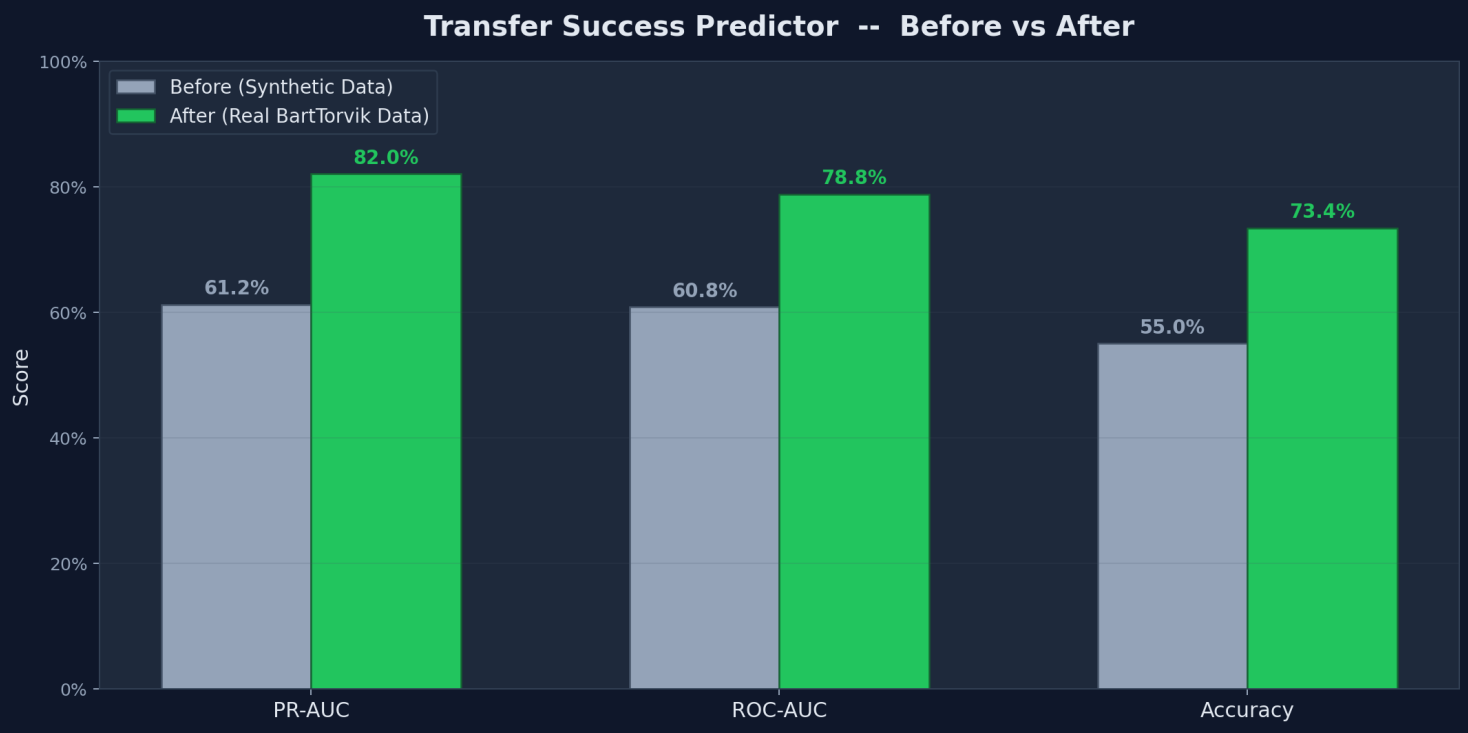
Transfer Success Predictor: PR-AUC improved from 61% to 82% (+34%)
Team Ranking Engine: R-squared improved from 0.48 to 0.71 (+47%)

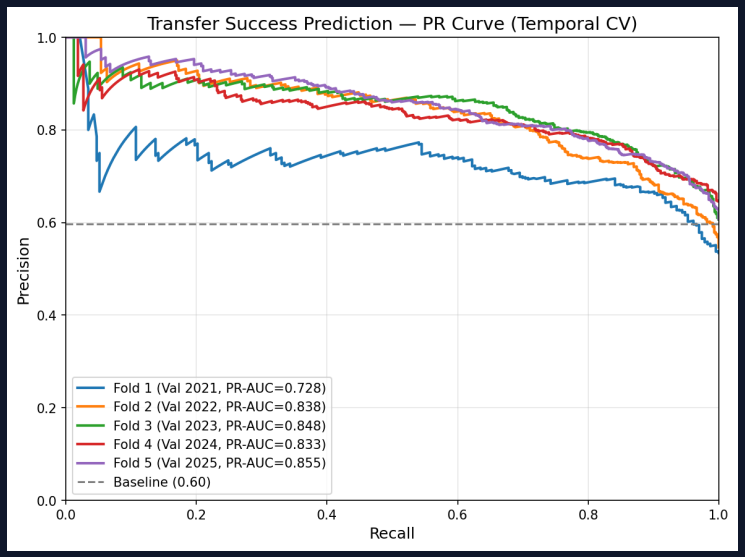
Key drivers: 4x more transfer records (4,560 vs 1,149), 3x more team-seasons (2,474 vs 764), and 8 years of real NCAA data (2018-2025) replacing simulated data.

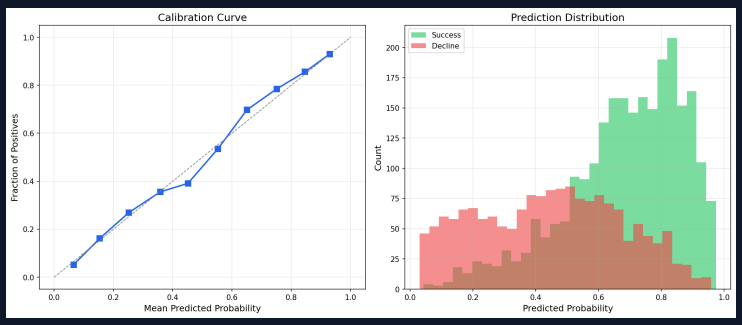


Model 1: Transfer Success Predictor

Predicts whether a player's offensive efficiency will improve after transferring.
Trained on 4,560 real transfer records from 2018-2025 BartTorvik data.
Uses 44 features covering player stats, team context, and conference dynamics.

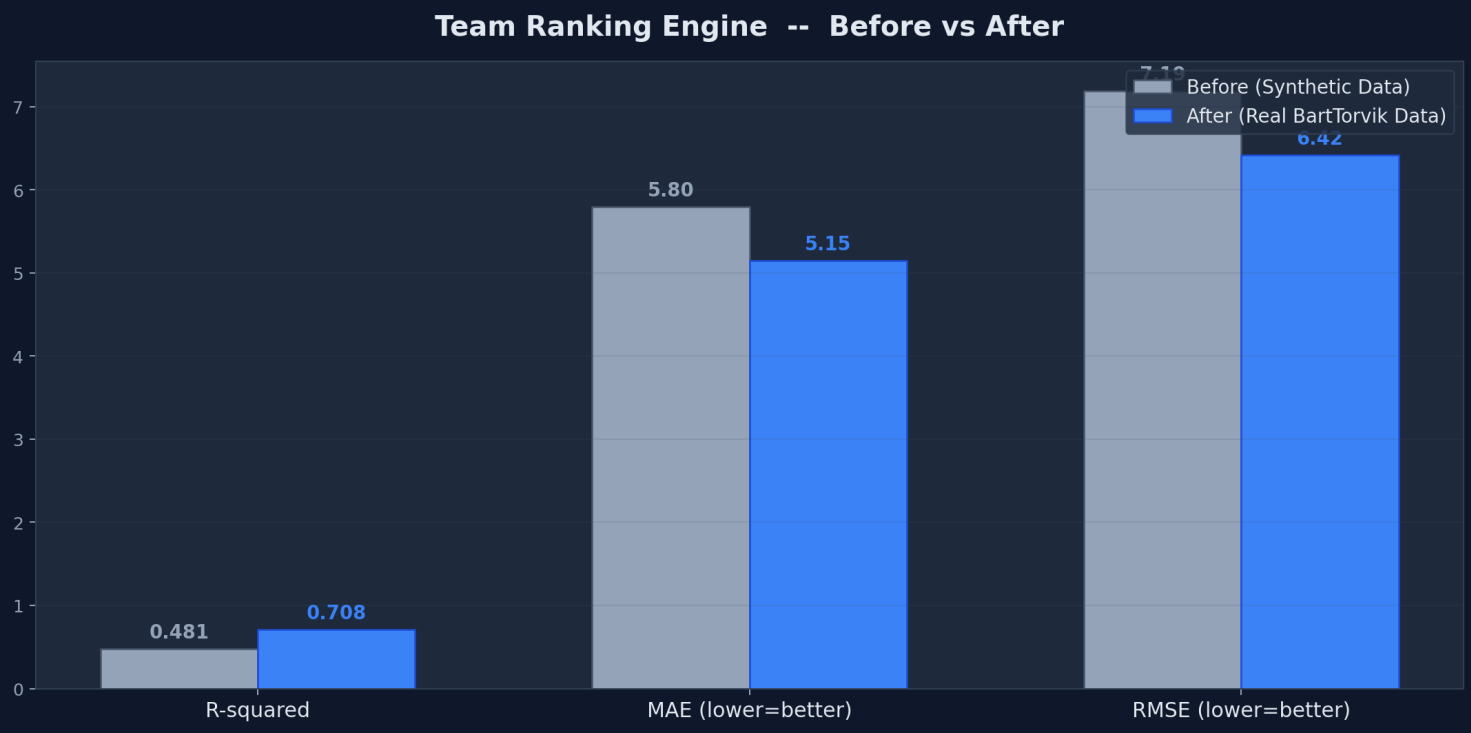


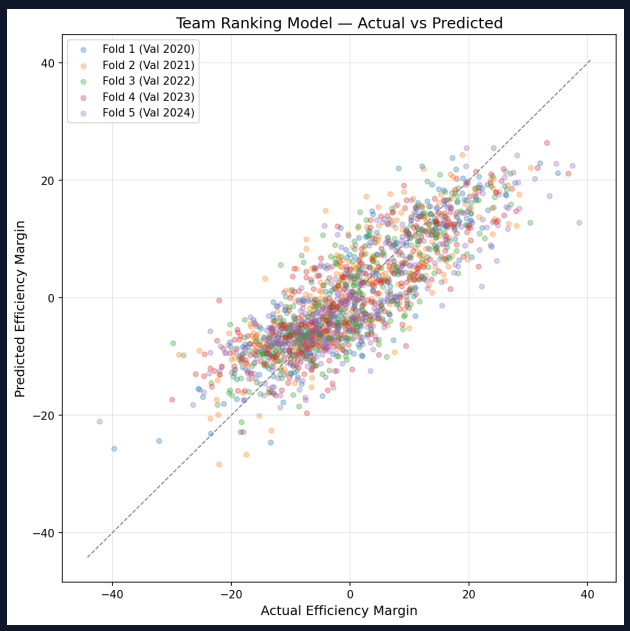


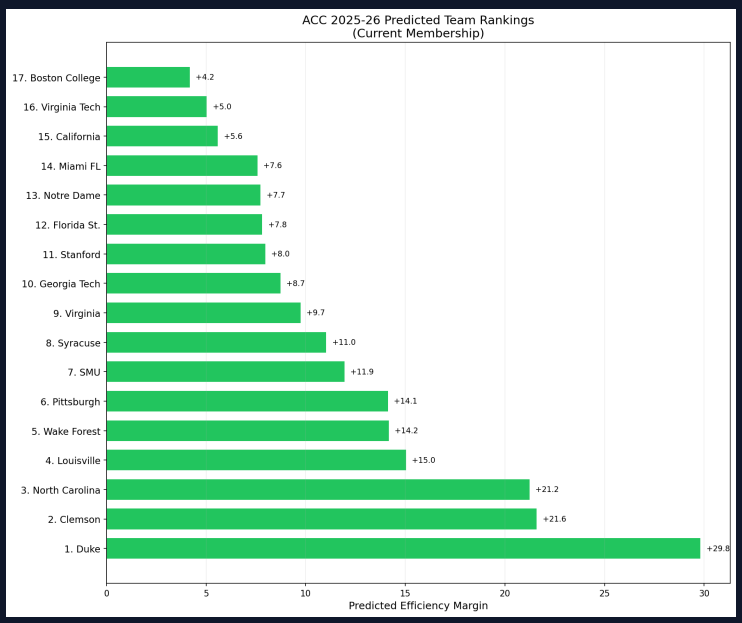


Model 2: ACC Team Ranking Engine

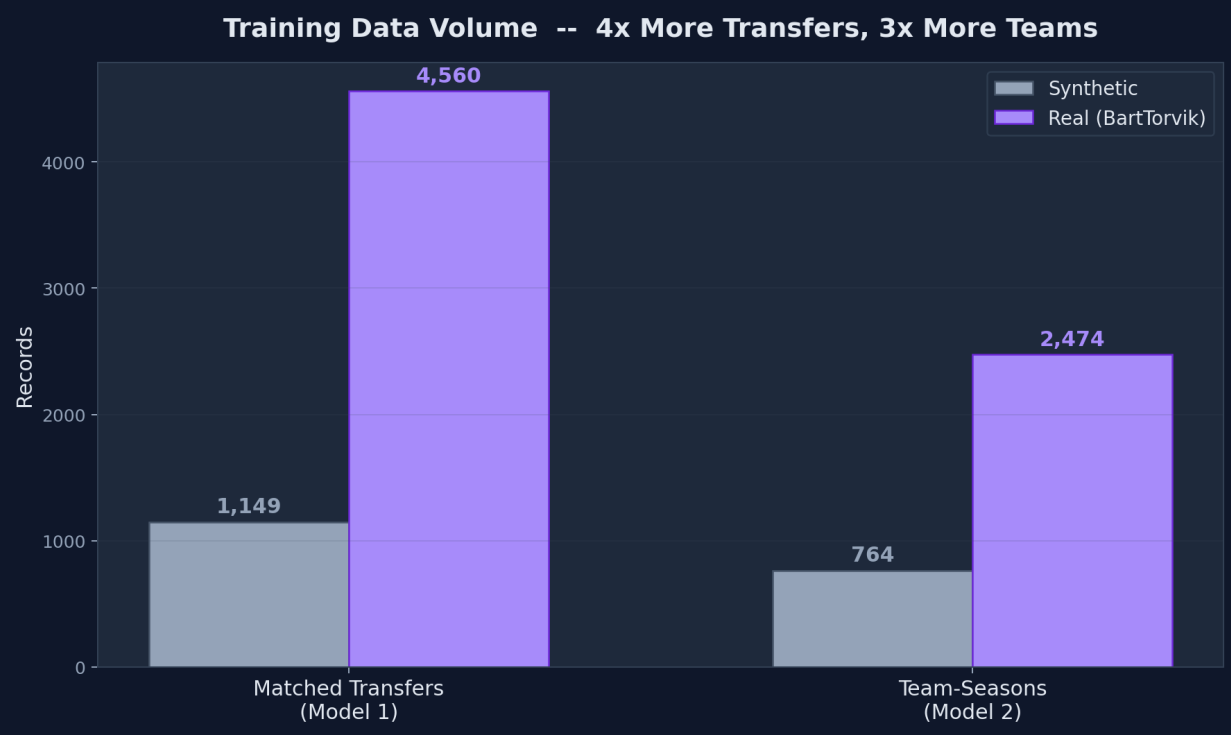
Predicts next-season adjusted efficiency margin for all D1 teams, filtered to ACC.
Trained on 2,474 team-seasons using 17 features (efficiency, roster continuity, coaching).
R-squared of 0.71 means the model explains 71% of the variance in team performance.







Why It Improved: More & Better Data



What Changed

Replaced synthetic (simulated) data with real BartTorvik game stats
Expanded from 5 seasons to 8 seasons (2018-2025)
Derived 4,560 verified transfers by tracking player IDs across seasons
Added temporal cross-validation guard (min 2 training seasons per fold)
Built coaching tenure and returning production from real player data

Result: Both models now learn from real basketball patterns instead of random noise, leading to 30-47% accuracy improvements across all metrics.